Kubernetes Node NotReady Resolution SOP

Objective

Restore cluster capacity when a Kubernetes node enters NotReady state due to kubelet, runtime, network, disk, or infrastructure failure.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Kubernetes cluster using kubectl and the correct namespace context.
2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T013
Issue	Kubernetes node not ready
Category	Infrastructure
Service	Kubernetes
Severity	Critical
Status	Resolved
Description	Node status showing NotReady
Expected Resolution	Check node logs and kubelet service

Procedure

1. Identify the NotReady node, affected pods, taints, and event timeline.
2. Run kubectl describe node to review conditions such as MemoryPressure, DiskPressure, PIDPressure, and NetworkUnavailable.
3. Check kubelet, container runtime, and node-level system logs.
4. Validate node CPU, memory, disk, network connectivity, and cloud provider health.
5. Cordon the node if workloads should not be scheduled there during investigation.
6. Drain the node only if required and approved for workload migration.
7. Restart kubelet or remediate infrastructure issue through the platform team.
8. Confirm node status returns Ready and workloads stabilize.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Node NotReady	Kubelet stopped	Restart kubelet and check logs
DiskPressure	Node disk full	Clean disk or increase capacity
NetworkUnavailable	CNI or node network issue	Check CNI plugin and routes
Pods evicted	Resource pressure	Scale capacity or tune requests

Best Practices

1. Alert on NodeNotReady and resource pressure.
2. Use pod disruption budgets for critical apps.
3. Keep node image and runtime versions maintained.
4. Review capacity before scheduling high-load workloads.

References

Kubernetes Docs | Internal Tickets CSV | Cluster Events | Monitoring Dashboards | Application Logs | Platform Runbook